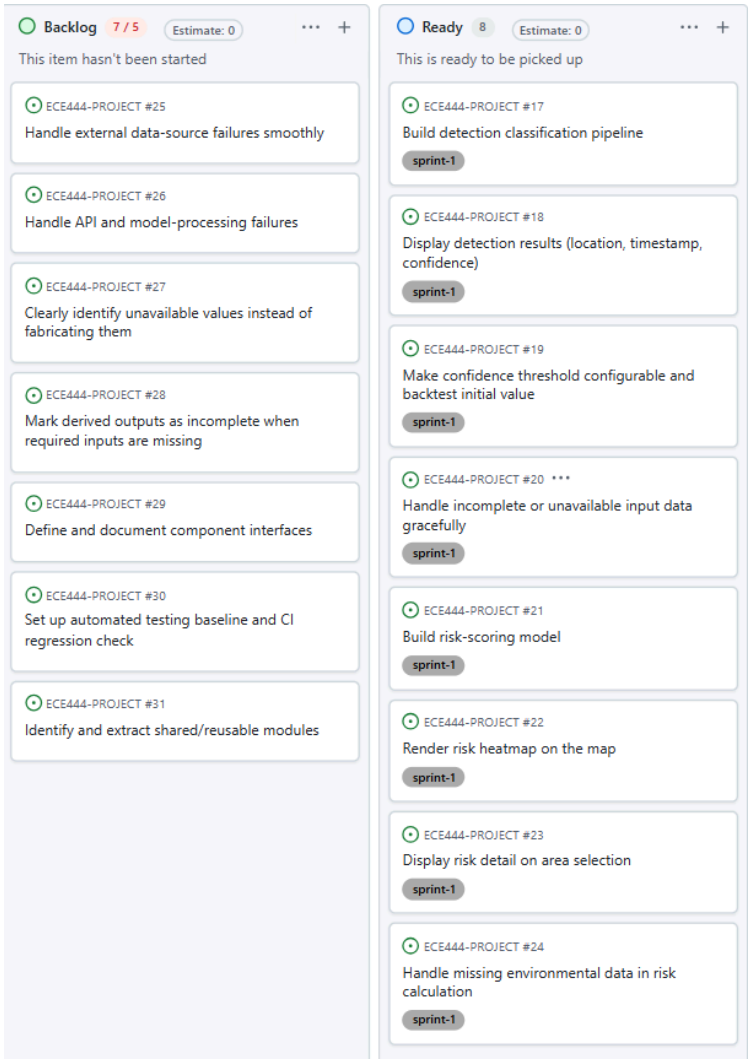
# Team Ash Detectives – Milestone 3.1: First Sprint Task Planning

Team Ash Detectives (wildfire early-detection and risk-monitoring platform) tracks work in GitHub Projects; Sprint 1 covers detection and risk heatmap, Sprint 2 covers reliability and maintainability.

## Project management tool and sorted backlog

Tool: **GitHub Projects** (Kanban board linked to our repository), with each implementation issue tracing back to a user story issue.

The backlog is sorted by priority. Detection (#10) and Heatmap (#3) sit in **Ready** for Sprint 1 because risk scoring and detection feed every other feature. Reliability (#4) and Maintainability (#12) stay in **Backlog** until there is working functionality to harden.



# Sprint 1 summary

Sprint 1 focused on the two stories every other feature depends on: **Wildfire Detection and Confidence Scoring (#10)** and **Wildfire Risk Heatmap (#3)**. Each story was split into 4 implementation issues (8 total), all moved to Ready on the GitHub Projects board.

- **Detection (#10):** classification pipeline using NASA FIRMS data, display of location/timestamp/confidence, configurable confidence threshold with an initial backtest, and graceful handling of missing input data.
- **Heatmap (#3):** risk-scoring model (#21), heatmap rendering on the map (#22), risk detail on area selection (#23), and handling of missing environmental data (#24).
- **Deferred to Backlog:** Reliability (#4, 4 issues) and Maintainability (#12, 3 issues), since they need working features to apply to.
- **Process:** pull requests are reviewed by a teammate before merging.

# Sprint 2 plan

Sprint 2 pulls the 7 backlog issues into scope and finishes anything that carried over from Sprint 1.

- **Reliability (#4, 4 issues):** external data-source failures (#25), API and model-processing failures (#26), clearly identified unavailable values (#27), and outputs marked incomplete when inputs are missing (#28).
- **Maintainability (#12, 3 issues):** defined and documented component interfaces (#29), automated testing baseline with a CI regression check (#30), and extraction of shared modules (#31).
- **Carry-over:** any unfinished Sprint 1 detection or heatmap issues.
- **Ongoing:** timely pull request reviews, team meetings, and test cases for the Sprint 1 stories.

# Tasks

**Mahrous's tasks for the 1st sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Build detection classification pipeline (#17) | 6 | 8 |

| | | |
|---|---|---|
| Display detection results: location, timestamp, confidence (#18) | 3 | 5 |
| Add unit tests for #17 and #18 | 2 | 2 |
| Review team-member pull requests timely and thoroughly | 2 | 2 |
| Participate actively in common team duties (e.g., meetings) | 2 | 2 |

**Mahrous's tasks for the 2nd sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Handle external data-source failures smoothly (#25) | 3 | |
| Handle API and model-processing failures (#26) | 3 | |
| Finish any carry-over Sprint 1 issues | 2 | |
| Add test cases for #25 and #26 | 2 | |
| Review team-member pull requests timely and thoroughly | 2 | |

| | | |
|---|---|---|
| Participate actively in common team duties (e.g., meetings) | 2 | |

**Malhotra's tasks for the 1st sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Render risk heatmap on the map (#22) | 5 | 7 |
| Handle incomplete or unavailable input data gracefully (#20) | 2 | 4 |
| Set up repository structure and coding standards as Software Lead | 2 | 2 |
| Review team-member pull requests timely and thoroughly | 2 | 2 |
| Participate actively in common team duties (e.g., meetings) | 2 | 2 |
| Write README file | 2 | 1 |

**Malhotra's tasks for the 2nd sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Define and document component interfaces (#29) | 3 | |
| Set up automated testing baseline and CI regression check (#30) | 4 | |
| Add test cases for #22 and #20 | 2 | |
| Review team-member pull requests timely and thoroughly | 3 | |
| Participate actively in common team duties (e.g., meetings) | 2 | |

**Cheralody's tasks for the 1st sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Display risk detail on area selection (#23) | 4 | 5 |
| Handle missing environmental data in risk calculation (#24) | 3 | 3 |

| | | |
|---|---|---|
| Sort backlog and maintain the GitHub Projects board | 2 | 2 |
| Write Contribution file | 2 | 2 |
| Review team-member pull requests timely and thoroughly | 2 | 2 |
| Participate actively in common team duties (e.g., meetings) | 3 | 2 |

**Cheralody's tasks for the 2nd sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Clearly identify unavailable values instead of fabricating them (#27) | 2 | |
| Identify and extract shared/reusable modules (#31) | 3 | |
| Update the board and write the sprint report | 3 | |
| Add test cases for #23 and #24 | 2 | |
| Review team-member pull requests timely and thoroughly | 2 | |

| | | |
|---|---|---|
| Participate actively in common team duties (e.g., meetings) | 3 | |

**Akbar's tasks for the 1st sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Make confidence threshold configurable and backtest initial value (#19) | 4 | 6 |
| Build risk-scoring model (#21) | 5 | 6 |
| Prepare and clean NASA FIRMS data for model input | 3 | 5 |
| Write Code of Conduct file | 2 | 1 |
| Review team-member pull requests timely and thoroughly | 2 | 2 |
| Participate actively in common team duties (e.g., meetings) | 2 | 2 |

**Akbar's tasks for the 2nd sprint**

| Task | Estimated Time (hour) | Actual Time (hour) |
|---|---|---|
| Mark derived outputs as incomplete when required inputs are missing (#28) | 2 | |
| Improve model using Sprint 1 evaluation results | 4 | |
| Add test cases for #19 and #21 (model integration) | 3 | |
| Review team-member pull requests timely and thoroughly | 2 | |
| Participate actively in common team duties (e.g., meetings) | 2 | |